



Herbal formula SYJN protect PC12 cells from neurotoxicity induced by corticosterone

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ABSTRACT

Aim of the study: SYJN is a Chinese herbal formula, containing four herbs: *Bupleurum chinense* DC., *Curcuma aromatica* Salisb., *Perilla frutescens* (Linn.) Britt. and *Acorus tatarinowii* Schott. Previous studies on the formula in our laboratory revealed an antidepressant-like effect on animal models of behavioral despair. However, the mechanisms underlying such antidepressant-like effect are yet to be understood. The aim of this work was to verify the previously established antidepressant-like effects on cell level using corticosterone-induced neurotoxicity in rat pheochromocytoma (PC12) cells to see if SYJN possesses any neuroprotective properties.

Materials and methods: PC12 cells were treated with 200 μ M corticosterone in the absence or the presence of various concentrations of SYJN for 48 h. Then, cell viability, apoptosis, intracellular Ca^{2+} ($[\text{Ca}^{2+}]_i$) concentration and caspase-3 activity were determined.

Results: Following the exposure of PC12 cells to 200 μ M corticosterone for 48 h, there were reductions in cell survival rate but increases in lactate dehydrogenase (LDH) release. In parallel, corticosterone caused significant elevations in DNA fragmentation, $[\text{Ca}^{2+}]_i$ concentration and caspase-3 activity. However, when the PC12 cells were incubated with SYJN at different concentrations (10, 50 and 100 mg/L) in the presence of 200 μ M corticosterone for 48 h, the above effects were evidently alleviated in a dose-dependent manner.

Conclusion: SYJN could generate a neuroprotective effect on corticosterone-induced neurotoxicity in PC12 cells, suggesting a possible action pathway of SYJN in vivo by decreasing the $[\text{Ca}^{2+}]_i$ concentration and caspase-3 activity.

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1. Introduction

Depression is a common, debilitating, life-threatening psychiatric disorder with a high prevalence. In spite of the introduction of various drugs such as the tricyclic antidepressants, monoamine oxidase inhibitor, selective serotonin reuptake inhibitor, noradrenergic reuptake inhibitor and noradrenaline reuptake inhibitor, depression continues to be a major medical problem. Consequently, searches for a safer, better-tolerated and powerful antidepressant drug continues (Yu et al., 2002; Bouvier et al., 2003).

The Chinese herbal formula SYJN has been successfully used to treat the depression-like disorders in clinical settings. It is composed of four herbal ingredients: *Bupleurum chinense* DC., *Curcuma aromatica* Salisb., *Perilla frutescens* (Linn.) Britt. and *Acorus tatarinowii* Schott.

The constitutive ratio of the four herbs is at 4:3:4:3 in dry weight. Previous pharmacological studies in our laboratory demonstrated that intragastric administration of SYJN significantly decreased immobility time in the tail suspension and forced swim tests in mice (Zhong et al., 2006a). We were also able to show that the behavioral alterations in unpredictable chronic stress treated mice and rats were reversed by chronic SYJN administration (Zhong et al., 2006b; Huang and Mao, 2007). However, the mechanisms by which SYJN exerts its antidepressant-like effects are yet to be elucidated.

On the other hand, the hippocampus is one of the brain structures that may be important for emotional and cognitive function. A recent MRI study showed that major depression was closely associated with reduced hippocampal volumes, and there was a positive correlation between the hippocampus atrophy and the time course of the depression (Li et al., 2003b; Saylam et al., 2006). Postmortem analyses indicated a reduction in the neuronal cells in hippocampus of patients with major depression (Stockmeier et al., 2004). Therefore, it is generally

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believed that neuronal atrophy and destruction in hippocampus have a causal role in the development and progress of depression (Manji and Duman, 2001; Fuchs et al., 2004). To date, the mechanism of neuronal degeneration in depression still remains to be uncovered, several lines of evidence support the involvement of the hyperactivation of hypothalamic–pituitary–adrenal (HPA) axis, with the latter being characterized by elevated levels of circulating glucocorticoids in blood (Magariños and McEwen, 1995; Sapolsky, 2000; Aihara et al., 2007; Murray et al., 2008).

For studying hippocampal neurons and the related neurons, neuronal cell lines are necessary. Rat pheochromocytoma (PC12) cell line, derived from rat pheochromocytoma tumors, is one of the widely used neuronal cell lines in a variety of studies. Treatment of PC12 cell line with high concentration of glucocorticoid to induce the neuronal damage in depressive disorder, has been used as an *in vitro* experimental model of depression (Li et al., 2003a,b, 2004; Zhu et al., 2006a,b; Gao et al., 2008). Different types of classical antidepressants have been shown to have protective effect against cytotoxicity induced by glucocorticoid in PC12 cells (Li et al., 2003b), suggesting that cytoprotection against glucocorticoid-induced neurotoxicity may represent one of the action pathways of antidepressants. So, in the present study, we used PC12 cells exposed to high concentrations of corticosterone, a principal glucocorticoid, to study the neuroprotective action of SYJN and the possible mechanisms.

2. Materials and methods

2.1. Preparation of SYJN

All the crude drugs of SYJN including 240 g of *Bupleurum chinense* DC., 180 g of *Curcuma aromatica* Salisb., 240 g of *Perilla frutescens* (Linn.) Britt. and 180 g of *Acorus tatarinowii* Schott. was purchased from Zhejiang Provincial Hospital of Traditional Chinese Medicine (Zhejiang Province, China). The voucher sample (no. 080321) has been deposited in College of Pharmacy, Zhejiang Chinese Medicine University. The mixture of SYJN was soaked in distilled water (1:8, w/v) for 0.5 h at room temperature with occasional stirring. They were prepared by boiling for 1 h, and the cooled decoction was filtered through two layers of cotton gauze. The procedure was repeated twice. The solution obtained was concentrated into residues in a vacuum evaporator, and lyophilized into powder. The yield of SYJN extract was 18.9 (w/w) according to the original herbs.

2.2. Materials

Corticosterone, DMEM culture medium and Fura-2/AM were purchased from Sigma–Aldrich (St Louis, MO, USA). Fetal bovine serum, heat-inactivated horse serum, penicillin and streptomycin were purchased from Gibco (Grand Island, NY, USA). All other chemicals and reagents were of analytical grade.

2.3. Cell culture and treatment

PC12 cells were obtained from the American Type Culture Collection (Rockville, MD, USA). PC12 cells were maintained in DMEM medium supplemented with penicillin (100 unit/ml), streptomycin (100 µg/ml), 5% fetal bovine serum and 5% horse serum at 37 °C in humidified atmosphere of 95% air and 5% CO₂. To study the neuroprotective effect of SYJN, PC12 cells were divided into five equal groups: non-treated control, 200 µM corticosterone and 200 µM corticosterone plus SYJN (10, 50 and 100 mg/L) in all experiments. Experiments were carried out 48 h after the cells were seeded.

After the media was removed, the cells were cultured in serum-free medium and incubated with corresponding drugs for another 48 h.

2.4. Cell viability assay

Cell survival was evaluated by a CellTiter 96® AQueous One Solution cell proliferation assay (Promega, Madison, WI). Briefly, PC12 cells were seeded on a 96-well culture plate at a density of 2×10^4 cells/well. At the end of the treatment, the media was removed. The cells were washed with D-Hanks, and then suspended in 100 µl of serum-free media. Subsequently, 20 µl of CellTiter 96® AQueous One Solution was added into each well of the 96-well assay plate and allowed to incubate for 1 h in a humidified atmosphere of 95% air with 5% CO₂ at 37 °C. The absorbance of each well was measured at 490 nm. Viability was expressed as percentage of non-treated control.

2.5. Lactate dehydrogenase (LDH) assay

LDH activity was measured using a LDH diagnostic kit (STANBIO Laboratory, USA) according to the manufacturer's protocol. Briefly, PC12 cells were seeded on 24-well culture plates at a density of 1×10^5 cells/well. At the end of the treatment, the medium was collected to measure the LDH activity. The absorbance of the samples was recorded at 340 nm at 1 min intervals for 3 min. The change in absorbance was then expressed in concentration units per liter. To determine intra-cellular LDH activity, the cells were washed by D-Hanks and then 500 µl D-Hanks were added to each well. The cells were lysed by ultrasonication (750 W) for 30 s. The suspension was centrifuged and supernatants were collected for assay of the LDH activity. LDH leakage was expressed as the percentage (%) of the total LDH activity (LDH in the medium + LDH in the cell), according to the equation % LDH released = (LDH activity in the medium/total LDH activity) $\times$ 100.

2.6. Quantification of DNA fragmentation

Quantification of DNA fragmentation was determined by Cell Death Detection ELISA^{plus} kit (Roche Applied Sciences, Basel, Switzerland) according to the manufacturer's protocol. Briefly, PC12 cells were seeded on a 96-well culture plate at a density of 2×10^4 cells/well. At the end of the treatment, the cells were resuspended and incubated in 200 µl of lysis buffer for 30 min at room temperature. After centrifuged to remove nuclei and cellular debris, 20 µl of the supernatant of each sample were transferred to a streptavidin-coated microplate, and incubated for 2 h at room temperature with an immunoreagent containing two monoclonal antibodies, anti-histone (biotin-labelled) and anti-DNA (peroxidase-conjugated). After washed three times with incubation buffer, each well was treated with the chromogen 2,2'-azinobis (3-ethylbenzthiazoline-6-sulfonic acid) as a substrate and then incubated for 15 min. The intensity of the color that develops was measured at 405 nm, while that at 490 nm was used as a blank (reference wavelength). Results were expressed as percentage of non-treated control.

2.7. Measurement of intracellular Ca²⁺ concentration

The concentration of [Ca²⁺]_i was determined as described previously elsewhere (Li et al., 2004; Zhu et al., 2006a,b). Briefly, PC12 cells were seeded on 24-well plates at a density of 1×10^5 cells/well. At the end of the treatment, the cells were collected and incubated with the complete medium containing 5 µM fura-2/AM at 37 °C for 45 min. Subsequently, the cells were washed and resuspended with cold balanced salt solution buffer containing 0.2% bovine serum albumin. The cells were incubated at 37 °C for another 5 min just

prior to measurement. The concentration of $[Ca^{2+}]_i$ was determined by alternating excitation wavelengths of between 340 and 380 nm with emission at 510 nm, using a fluorescence spectrophotometer (F-4500, HITACHI, Japan). Results were expressed as percentage of non-treated control.

2.8. Measurement of caspase-3 activity

The activity of caspase-3 like protease was measured using a colorimetric caspase-3 assay kit (Sigma–Aldrich, St Louis, MO, USA) according to the manufacturer's protocol. Briefly, PC12 cells were seeded on 6-well culture plates at a density of 2×10^6 cells/well. At the end of the treatment, the media was removed, and then 500 μ l of lysis buffer were added to each well. The cells were incubated in lysis buffer on ice for 30 min. The lysates were centrifuged and supernatants were collected. The reaction mixture (total volume, 100 μ l) containing 30 μ l of supernatant and 10 μ l of caspase-3 substrate acetyl-Asp-Glu-Val-Asp-p-nitroanilide (final concentration, 200 μ M) in assay buffer was added to a 96-well plate, and incubated at 37 °C for 90 min. The absorbance of each well was measured at 405 nm. Results were expressed as percentage of non-treated control.

2.9. Statistical analysis

Data were expressed as mean $\pm$ S.E.M. One-way analysis of variance (ANOVA) followed by Dunnett's test was used to test the differences between the groups. The GraphPad Prism software was used to perform the statistics (version 4.0; GraphPad Software, Inc., San Diego, CA). A difference was considered statistically significant when the value $p < 0.05$.

3. Results

3.1. Effect of SYJN on the cell viability in corticosterone-treated PC12 cells

The effect of SYJN on the cell viability in corticosterone-treated PC12 cells was shown in Fig. 1. The viability of PC12 cells when exposed to corticosterone at concentration of 200 μ M for 48 h was significantly decreased as compared with the control group ($F(4, 30) = 126.1$, $p < 0.01$) and the survival rate was 40.9% of control. When PC12 cells were treated with various concentrations of SYJN (10, 50 and 100 mg/L) in the presence of 200 μ M corticosterone for 48 h, the cell viability was significantly increased ($F(4, 30) = 126.1$, $p < 0.05$, $p < 0.01$ and $p < 0.01$ respectively) as compared with the corticosterone group, and the survival rate was 47.7%, 55.8% and 61.0% of control respectively.

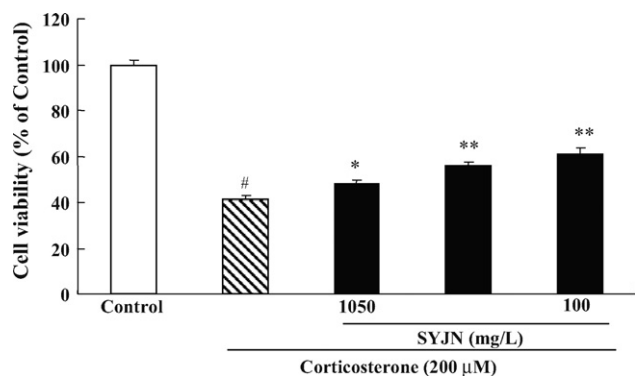


Fig. 1. Effect of SYJN on the cell viability in corticosterone-treated PC12 cells. Values given are the mean $\pm$ S.E.M. ($n = 7$). # $p < 0.01$ as compared with control group; * $p < 0.05$, ** $p < 0.01$ as compared with corticosterone group.

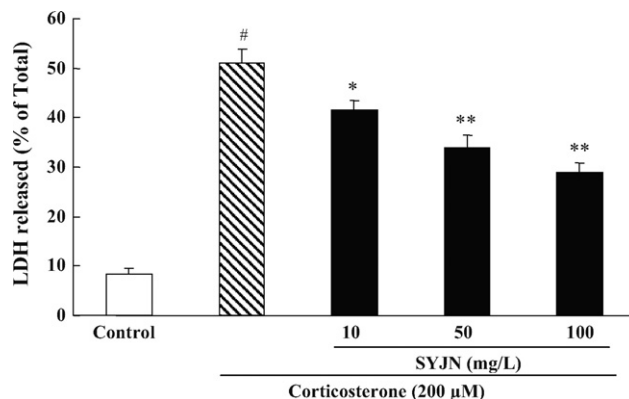


Fig. 2. Effect of SYJN on the LDH leakage in corticosterone-treated PC12 cells. Values given are the mean $\pm$ S.E.M. ($n = 6$). # $p < 0.01$ as compared with control group; * $p < 0.05$, ** $p < 0.01$ as compared with corticosterone group.

3.2. Effect of SYJN on the LDH leakage in corticosterone-treated PC12 cells

The effect of SYJN on the LDH leakage in corticosterone-treated PC12 cells was shown in Fig. 2. After treatment of PC12 cells with 200 μ M corticosterone for 48 h, the LDH leakage was significantly increased as compared with the control group ($F(4, 25) = 54.3$, $p < 0.01$) and the percentage of LDH leakage in corticosterone group was 51.2% while 8.4% in control group. Co-treatment of PC12 cells with various concentrations of SYJN (10, 50 and 100 mg/L) in the presence of 200 μ M corticosterone for 48 h could result in the decrease of LDH leakage ($F(4, 25) = 54.3$, $p < 0.05$, $p < 0.01$ and $p < 0.01$ respectively), and the percentage of LDH leakage was 41.4%, 34.0% and 28.8% respectively.

3.3. Effect of SYJN on the DNA fragmentation in corticosterone-treated PC12 cells

The effect of SYJN on the DNA fragmentation in corticosterone-treated PC12 cells was shown in Fig. 3. After treatment of PC12 cells with 200 μ M corticosterone for 48 h, the DNA fragmentation was significantly increased as compared with the control group ($F(4, 25) = 133.0$, $p < 0.01$) and the extent of DNA fragmentation was 334.9%, while that of 100% in the control group. When PC12 cells were treated with various concentrations of SYJN (10, 50 and 100 mg/L) in the presence of 200 μ M corticosterone for 48 h, the DNA fragmentation was significantly decreased ($F(4, 25) = 133.0$, $p < 0.05$, $p < 0.01$ and $p < 0.01$ respectively) as compared with the corticosterone group, and the extent of DNA fragmentation was 309.0%,

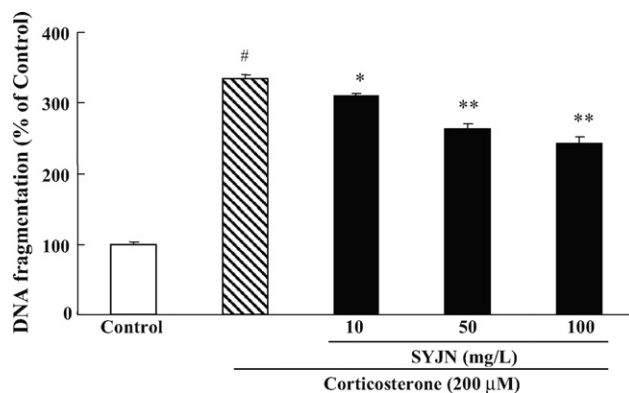


Fig. 3. Effect of SYJN on the DNA fragmentation in corticosterone-treated PC12 cells. Values given are the mean $\pm$ S.E.M. ($n = 6$). # $p < 0.01$ as compared with control group; * $p < 0.05$, ** $p < 0.01$ as compared with corticosterone group.

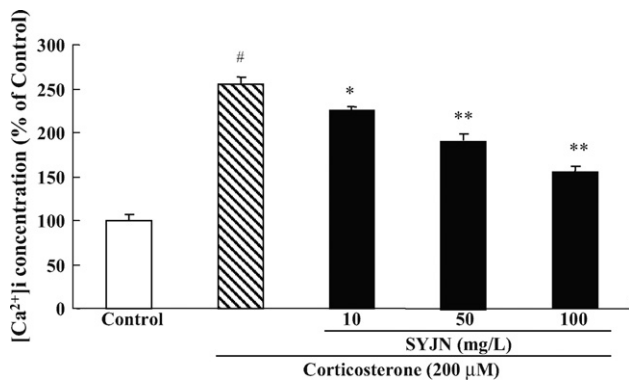


Fig. 4. Effect of SYJN on the concentration of $[Ca^{2+}]_i$ in corticosterone-treated PC12 cells. Values given are the mean $\pm$ S.E.M. ($n=6$). # $p<0.01$ as compared with control group; * $p<0.05$, ** $p<0.01$ as compared with corticosterone group.

263.3% and 242.3% respectively.

3.4. Effect of SYJN on the concentration of $[Ca^{2+}]_i$ in corticosterone-treated PC12 cells

The effect of SYJN on the concentration of $[Ca^{2+}]_i$ in corticosterone-treated PC12 cells was shown in Fig. 4. After treatment of PC12 cells with 200 μ M corticosterone for 48 h, the concentration of $[Ca^{2+}]_i$ was significantly increased as compared with the control group ($F(4, 25)=79.2$, $p<0.01$) and the $[Ca^{2+}]_i$ concentration was 255.6%, while that of 100% in the control group. When PC12 cells were treated with various concentrations of SYJN (10, 50 and 100 mg/L) in the presence of 200 μ M corticosterone for 48 h, the concentration of $[Ca^{2+}]_i$ was significantly decreased ($F(4, 25)=79.2$, $p<0.05$, $p<0.01$ and $p<0.01$ respectively) as compared with the corticosterone group, and the $[Ca^{2+}]_i$ concentration was 224.7%, 191.8% and 154.9% respectively.

3.5. Effect of SYJN on the activity of caspase-3 in corticosterone-treated PC12 cells

The effect of SYJN on the activity of caspase-3 in corticosterone-treated PC12 cells was shown in Fig. 5. After treatment of PC12 cells with 200 μ M corticosterone for 48 h, the activity of caspase-3 was significantly increased as compared with the control group ($F(4, 25)=207.0$, $p<0.01$) and the caspase-3 activity was 308.2%, while that of 100% in the control group. When PC12 cells were treated with various concentrations of SYJN (10, 50 and 100 mg/L) in the presence of 200 μ M corticosterone for 48 h, the activity of caspase-3 was

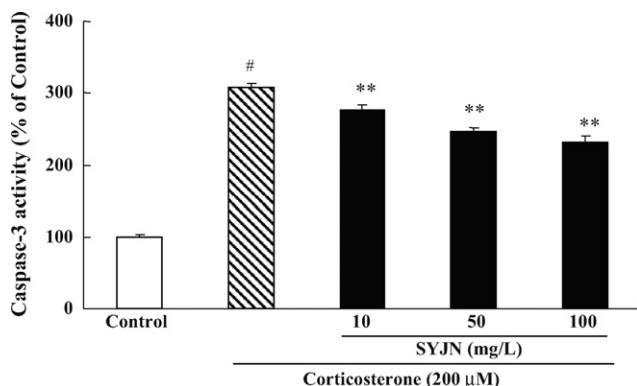


Fig. 5. Effect of SYJN on the activity of caspase-3 in corticosterone-treated PC12 cells. Values given are the mean $\pm$ S.E.M. ($n=6$). # $p<0.01$ as compared with control group; * $p<0.05$, ** $p<0.01$ as compared with corticosterone group.

significantly decreased ($F(4, 25)=207.0$, $p<0.01$, $p<0.01$ and $p<0.01$ respectively) as compared with the corticosterone group, and the caspase-3 activity was 276.0%, 245.2% and 230.3% respectively.

4. Discussion

In the present study, we found that corticosterone caused a significant decrease in the cell viability and increase in the LDH leakage, confirming its neurotoxicity in PC12 cells. In this *in vitro* model, SYJN partly reversed these changes induced by corticosterone, supporting the antidepressant-like effect of SYJN as we reported *in vivo*. Consistently, animal studies demonstrated that abnormally high corticosteroid levels could induce depression like behaviours (Johnson et al., 2006; Murray et al., 2008) and cause many functional changes in hippocampal neurons (Magariños and McEwen, 1995; Sapolsky, 2000; Murray et al., 2008). Furthermore, *in vitro* studies reported that high concentration of corticosterone induced damage in primary cultured hippocampal neurons as well as in the cultured PC12 cells (Li et al., 2003b, 2004, 2007a,b; Zhu et al., 2006a,b).

Importantly, we found that SYJN suppressed the corticosterone-induced apoptosis in PC12 cells. Apoptosis is involved in the pathological changes in the hippocampus of patients with depression (Lucassen et al., 2001). Our previous studies and other reports also confirmed that the number of apoptotic cells in the hippocampus was increased in chronic mild stress model of depression of rats, and could be reduced by antidepressant treatment (Lucassen et al., 2004; Zhong et al., 2006c). Apoptosis in hippocampal neurons might therefore be one of the pathogenetic factors involved in the development of experimental and clinical depression (Arantes-Gonçalves and Coelho, 2006; Lucassen et al., 2006; Kosten et al., 2008).

Moreover, in the present study, SYJN significantly reversed the corticosterone-enhanced $[Ca^{2+}]_i$ concentration and reduced the increased caspase-3 activity in PC12 cells. $[Ca^{2+}]_i$ plays a key role in the development of neurons and $[Ca^{2+}]_i$ overloading may result in neuronal injury (Li et al., 2003a, 2004; Zhu et al., 2006a,b). Several studies reported that corticosterone-induced apoptotic cell death in PC12 cells was related to $[Ca^{2+}]_i$ overloading (Li et al., 2004; Zhu et al., 2006a,b). On the other hand, Ca^{2+} -independent cysteine proteases is also involved in corticosterone induced-neurotoxicity including cell apoptosis (Li et al., 2007a,b; Zhou et al., 2008). Caspases are a family of cysteine proteases which play crucial roles in the complex biochemical signaling events in triggering apoptosis. (Li et al., 2007a,b). Recently, among 14 known members of the caspase family of proteases, caspase-3 was believed to be a frequently activated death protease, catalyzing the specific cleavage of many key cellular proteins (Cheung et al., 2008). Previous report suggested that corticosterone-induced apoptosis in primary cultured hippocampal neurons was related to the increase of caspase-3 activity (Zhu et al., 2006a,b). In the current study, caspase-3 activity in PC12 cells was significantly enhanced by corticosterone but significantly decreased by SYJN. These results suggested that the neuroprotective effects of SYJN in PC12 cells might be mediated, at least in part, by the inhibition of $[Ca^{2+}]_i$ overloading and caspase-3 activity.

In summary, our results show that SYJN can inhibit the corticosterone-induced apoptosis in PC12 cells through decreasing the $[Ca^{2+}]_i$ concentration and caspase-3 activity, and which may be one of the most important mechanisms of its antidepressant effects.

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